

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Mid-Term Examination, 2025-26 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-43 / MATMN-41 : Differential Equations
(Mid-Term)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Attempt any FIVE questions. All questions carry equal marks (4 marks each).

Question 1. Solve Clairaut's differential equation $y = px + f(p)$ where $p = \frac{dy}{dx}$, and obtain its general and singular solutions. [4]

Question 2. Solve the second-order linear ODE $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2e^{2x}$. [4]

Question 3. Solve by the Method of Variation of Parameters: $\frac{d^2y}{dx^2} + y = \sec x$. [4]

Question 4. Solve Cauchy-Euler homogeneous linear differential equation $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = x^2 \ln x$. [4]

Question 5. Find the orthogonal trajectories of the family of cardioids $r = a(1 + \cos \theta)$. [4]

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